State of Charge Estimation and State of Health Diagnostic Method Using Multilayer Neural Networks

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***Abstract*— Lithium batteries are the most common energy storage devices in fields such as electric vehicles, portable devices, and energy storage systems. Continuously using the battery in a degradation state creates a fire or explosion risk. To prevent such accidents, research on a battery management system (BMS) that diagnoses the state of a battery was conducted. This study proposes a method that uses multilayer neural networks (MNN) for state of charge (SOC) estimation and state of health (SOH) diagnosis. The proposed method uses four MNN models as SOH diagnostic models and three SOC estimation models. Each SOC estimation model comprises a normal model, a caution model, and a fault model according to the learned data based on the output result of the SOH diagnostic model. From the experiments, the proposed method estimates and diagnoses SOC and SOH well.**

***Keywords*— lithium battery; state of health; state of charge; multilayer deep neural network; estimation method**

I. INTRODUCTION

Lithium batteries are the most common energy storage devices in fields such as electric vehicles, portable devices, and energy storage systems (ESS) [1–3]. Battery performance when connected to a load or source is based on the chemical reactions inside the battery. The chemicals degrade with time and usage, reflecting a gradual decline in the energy storage capacity of the battery [4]. Several environmental factors affect battery life, such as operating at high and low temperatures, frequent charging, and over-discharging. If the battery is continuously used in a degradation state, there is a risk of fire or explosion. To prevent such accidents, research on a battery management system (BMS) that diagnoses the state of a battery was conducted.

State of charge (SOC) and state of health (SOH) are parameters required for battery diagnosis or estimation. The SOC refers to the remaining capacity in the battery, and by estimating SOC, it is possible to know the usage time of the battery. It prevents over-discharging and overcharging of the battery, enabling the stable use of the battery. The SOH refers to the state of the battery, and diagnosing SOH can evaluate the replacement timing of the battery. Therefore, it is an efficient operating system because it prevents accidents such as fire and explosion because of the progress of battery aging. The battery voltage, current, temperature, and operating environment of the battery affect SOC and SOH; therefore, an algorithm that can model the characteristics of these batteries is required.

Several methods exist to estimate the SOC, such as the current integral and open-circuit voltage (OCV)-SOC curve methods using the relationship between OCV and SOC. The current integration method measures current in real-time. Although this method uses simple calculations and is intuitive, it requires knowing the battery's initial SOC and excludes variables such as sensor error, battery age, and external temperature [5]. The OCV-SOC curve-based method requires OCV measurements. It is impossible to measure the OCV in real-time because the measurement must be taken after the battery has rested for 2 h or more, while the circuit is open.

SOH diagnostic methods can use a Kalman-filter approach. This method estimates the resistance of a battery's equivalent circuit [6] but is challenging to apply because of the complexity of its parameters and algorithms. Today, neural networks are applied to problems such as pattern recognition, identification, and classification and are promising possibilities for BMS solutions [7–8].

This study proposes a method using multilayer neural networks (MNN) for SOC estimation and SOH diagnosis. The proposed method uses four MNN models as SOH diagnostic models and three SOC estimation models. Each SOC estimation model comprises a normal model, a caution model, and a fault model according to the learned data. Based on the output result of the SOH diagnostic model, the result of the SOC model for the battery state is the output. For example, if the output of the SOH diagnostic model is normal, the SOC estimation result of the normal model is the output. From the experiments, the proposed method estimates and diagnoses SOC and SOH well.

II. PROPSED METHOD

A. State of Charge

The SOC refers to the battery's remaining current capacity and is a critical parameter to consider when monitoring the battery. The SOC is a percentage of the current releasable capacity from the battery's rated capacity, as shown in Eq. (1).

$$SOC = \frac{C_{releasable}}{C_{rated}} \times 100 \tag{1}$$

The SOC can be calculated in the coulomb counter. However, this method has the disadvantage that the correct SOC value cannot be calculated if the initial SOC setting is incorrect or a sensor error is accumulated. This study estimated the SOC using a neural network that can model battery characteristics.

B. State of Health

The SOH is an indicator of performance degradation because of the battery, and the battery state can be measured based on the SOH. The SOH refers to the current battery capacity as a percentage from the initial capacity, as shown in Eq. (2).

$$SOH = \frac{C_{actual}}{C_{initial}} \times 100 \tag{2}$$

If the capacity of the battery decreases to lower than 80% of its initial capacity, it is a failure [9]. In this study, when the SOH is 100%–90%, it is in a normal state, when it is 90%–80%, it is in a caution state, and when it is lower than 80%, it is defined as a fault state.

C. Multilayer Neural Network

The MNN is the most basic learning model in multilayer learning and refers to a model with two or more hidden layers (Fig. 1). It allows high abstraction levels by combining various nonlinear transform techniques with more hidden layers than found in existing neural networks.

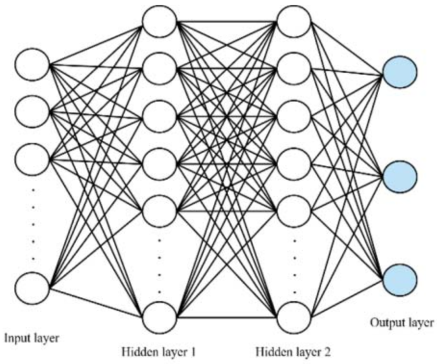

Fig. 1. Structure of a multilayer neural network

The learning optimization method was RMSprop for the SOC and SOH models. RMSprop is governed by Eq. (3) or (4).

$$G_t = \gamma G + (1 - \gamma)(\nabla_\theta f_t(\theta_t))^2 \tag{3}$$

$$\theta_t = \theta_{t-1} - \sqrt{\frac{\eta}{G_t - \epsilon}} \cdot \nabla_\theta f_t(\theta_{t-1}) \tag{4}$$

where γ is the decay rate (typically 0.9), θ is a network parameter, and $f(\theta)$ is the cost function. This study uses softmax. $\nabla_\theta f(\theta)$ is the gradient of the network, and t is a time step. G_t is an accumulated squared gradient, and η is the learning rate that is set to 0.001. ϵ refers to epsilon.

The activation function for the MNN nodes was the rectified linear unit (ReLU) function as defined in Eq. (5).

$$f(x) = \begin{cases} x, for\ x > 0 \\ 0, for\ x \leq 0 \end{cases} \tag{5}$$

The softmax function is used for the output layer and to transform the class classification problem. The output from the previous layer is input into the probability of each state when solving the state classification problem. The softmax function is as in Eq. (6).

$$p_i = \frac{\exp(x_i)}{\sum_k \exp(x_k)} \tag{6}$$

where k is the number of dimensions of the input and output vectors. p_i is the i-th output value, and x_i is the i-th input value.

D. Proposed SOC Estimation and SOH Diagnostic Method

The existing SOC estimation method uses parameters such as voltage, current, temperature, and internal resistance to model the deterioration characteristics of a battery to estimate the SOC [10–12]. The proposed method diagnoses one of three states, namely normal, caution, and fault through the threshold defining the battery SOH. From the diagnostic result, the SOC estimation result of the MNN model corresponding to the battery state is selected and output (Fig. 2).

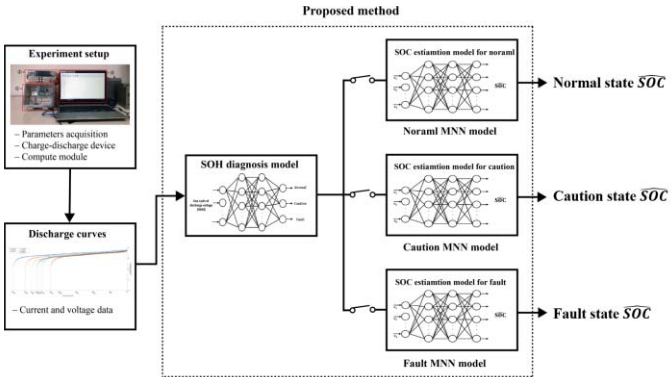

Fig. 2. Structure of proposed method

The proposed method uses four MNN models. One is an MNN model for SOH diagnosis, and the other three are models for SOC estimation. The SOC estimation models model the normal data set with 100%–90% SOH, the caution data set with 90%–80%, and the fault state data set with less than 80%. Using the learned data, it is defined as a normal model, a caution model, and a fault model, respectively.

The proposed method process is described below. First, the battery voltage data are input from the experimental device. From the data classification, a discharge voltage data set of one cycle and a discharge voltage data set of six inputs are generated. The discharge voltage data set for one cycle is input into the SOH diagnostic model and classified into normal, caution, or fault states. The discharge voltage data of six inputs are input into each of the 3 SOC estimation models to estimate SOC.

From the result of the SOH diagnostic model, one result of the three SOC estimation models is selected and output. If the SOH diagnosis result is normal, the SOC estimation result of the normal model is output, and if the diagnostic result is caution, the caution model is output. Finally, if the two conditions are not met, the fault model is output. Fig. 3 shows this method in detail through the flow chart.

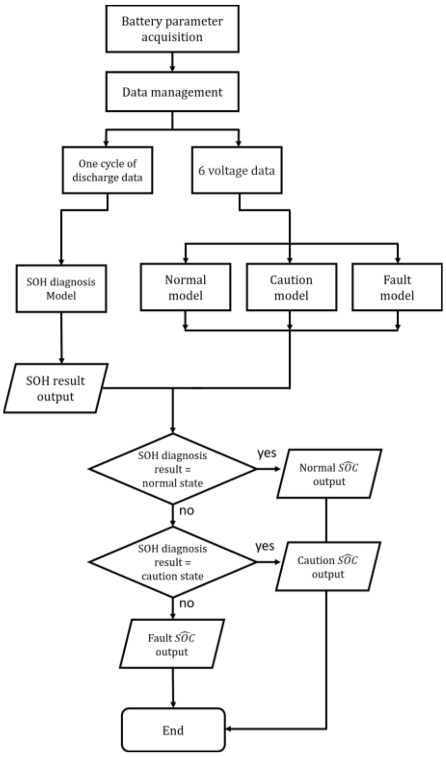

Fig. 3. Flow chart of proposed method

III. EXPERIMENT AND RESULTS

The performance of the proposed system was evaluated using the discharge voltage data of the battery operated as an experiment. Fig. 4 shows the training and test data.

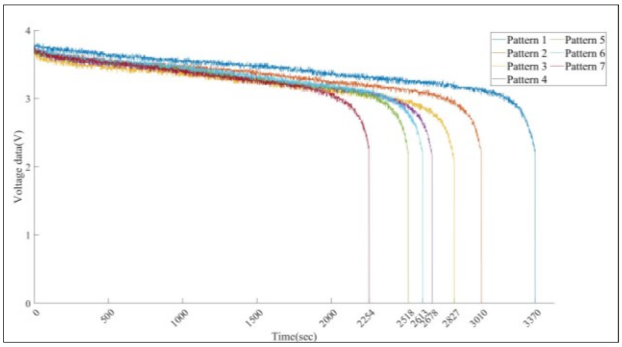

Fig. 4. Battery discharge voltage data

Pattern 1 is a normal state with 100% of the rated capacity. Patterns 2–4 are in the caution state with 90%–80% of the rated capacity. Patterns 5–7 are in the fault state with less than 80% rated capacity. Pattern 1 was used as training data for the MNN of the normal model, patterns 2–4 were used as caution models, and patterns 5–7 were used for fault models.

The proposed SOC and SOH estimation and diagnostic methods are based on MNN as described below. Fig. 5 shows the MNN models for SOC estimation.

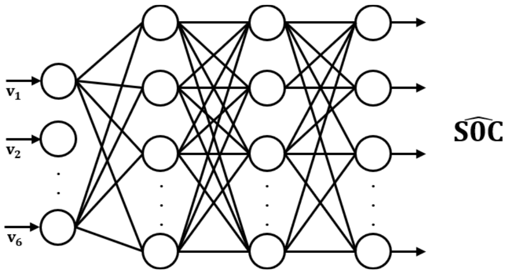

Fig. 5. MNN structure for SOC estimation

The MNN models for SOC estimation are in a 6-150-150-101 MNN structure (6 input nodes, 150 hidden nodes in hidden layer 1, 150 hidden nodes in hidden layer 2, and 101 output nodes). The network was trained with RMSprop optimization, and each hidden layer was activated using the ReLU function, and the output layer used softmax.

Fig 6. shows the MNN models for SOH diagnosis. 6. The MNN models for SOC estimation are in a 3600-256-256-3 MNN structure. The learning method and the activation function used are the same as for the SOC model.

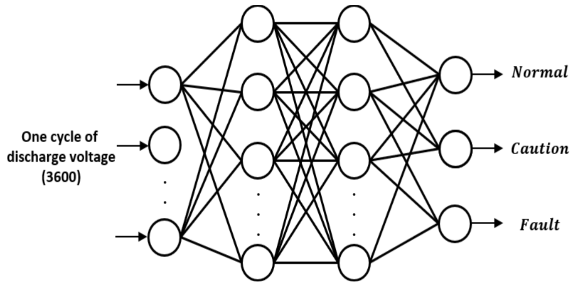

Fig. 6. MNN structure for SOH diagnosis

The data of pattern 1, pattern 2, and pattern 6 were used for the test. The SOC and SOH test results are set out below.

Fig. 7 shows the SOH diagnosis results. Each pattern was accurately diagnosed into normal, warning, or fault states.

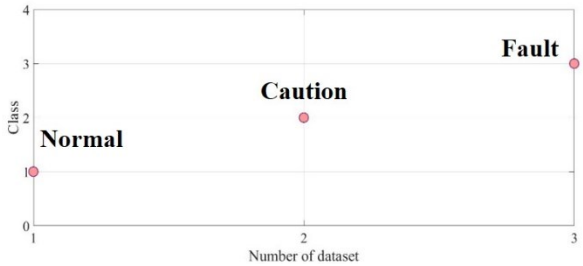

Fig. 7. SOH diagnosis result.

Table 1 shows the SOC estimation results. The normal model learns pattern 1 data, and the caution model learns pattern 2 to pattern 4. The fault model learns pattern 5 to pattern 7 data, and the total model learns pattern 1 to pattern 7 data. The SOC estimation result was verified with a mean absolute error.

TABLE I. SOC ESTIMATION RESULTS

	Pattern 1	Pattern 2	Pattern 6
Normal model	0.01%	9.07%	10.11%
Caution model	13.3%	2.03%	6.01%
Fault model	8.82%	2.17%	0.1%
Total model	7.12%	2.51%	2.29%

Pattern 1 shows the highest performance at 0.01% in the normal model. Pattern 2 shows the highest performance at 2.03% in the caution model. Pattern 6 shows a 0.1% highest performance in the fault model. It shows 7.11%, 0.48%, and 2,19% lower error than the total model that learned all the patterns.

IV. CONCLUSION

This study proposed a SOC estimation and SOH diagnostic method using MNN. The battery was charged and discharged in a high-temperature chamber (40 °C), and the battery discharge voltage data obtained in the experiment were used to estimate the SOH and SOC of the battery using the proposed method. The results show that the SOH diagnostic model diagnosed the correct state of the battery, and the SOC estimation model showed that the SOC estimation performance of the model using the proposed method was better than the model learning all the data. In the future, we will apply it to systems to verify the performance of the method in real environments.

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